

Lab 2 – Pandas and data wrangling

DAVE3625 Introduction to Artificial Intelligence · Høst 2026

The `stud.csv` DataFrame has 50 entries with: StudentID, Age, email, hrsStudy, FinalGrade

Tasks

1. In this lab, you will import the csv file into pandas:

Hint:

```
#If you want to use the csv from this git set
# url = "https://raw.githubusercontent.com/DAVE3625/Dave3625-Host-2025/refs/heads/main/Lab2/data/stud.csv"
# You can also download the csv and set
# url="{filepath}/stud.csv"
df = pd.read_csv(url, sep=',')
df.head()
```

2. You will then clean the data set so `df.info()` produce

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 48 entries, 0 to 49
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   StudentID   48 non-null    int64
 1   Age         48 non-null    int32
 2   email       48 non-null    object
 3   hrsStudy    48 non-null    int32
 4   FinalGrade  48 non-null    float64
```

Hint:

```
df.isna().sum() #show missing values
df=df.replace(r'^\s*$', np.nan, regex=True) #Replace blank values with np.nan values

df['Column'] = df['Column'].astype(str).astype(int) #Convert from obj to int
```

3. Then identify and remove the outliers in the «FinalGrade» column

Hint :

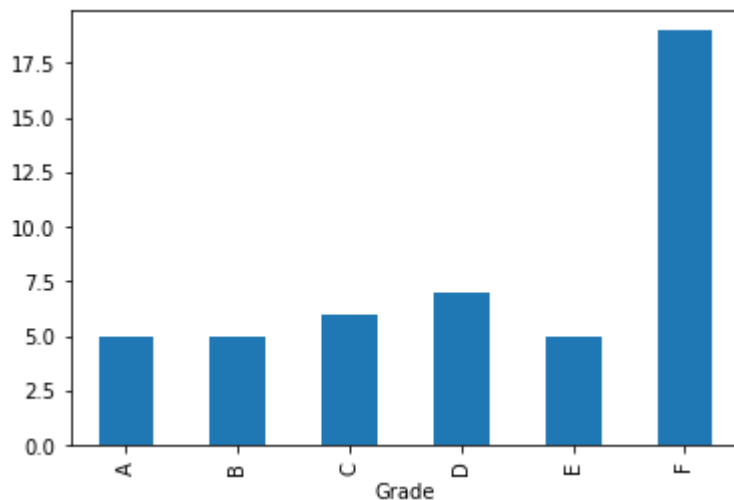
```
df["FinalGrade"].plot.box()
```

4. Finally add a column “Grade” where you transform the grade from float to a char:

Hint

```
91 - 100 = A
81 - 90  = B
71 - 80  = C
61 - 70  = D
51 - 60  = E
> 50    = F
```

5. Produce this plot:



The dataset is generated with this script and errors added after it's creation:

```
import random
print(StudentID, Age, email, hrsStudy, FinalGrade)
for i in range(50):
    studId = "s" + str(random.randrange(10000, 99999, 1))
    print(str(studId) + ", " + str(random.randrange(19, 35, 1)) + ", "
          + str(studId + "@oslomet.no") + ", " + str(random.randint(0, 12)) + ", " + str(random.randint(20, 100)))
```

More hints

Due to many questions about pandas and python in general i'll provide some extra hints

Q: I get an error when running

```
df = pd.read_csv(url, sep=',')
```

A: This might be caused by two different problems

1. Check your imports, for this lab you should consider these imports

```
#Import modules
%matplotlib inline
import pandas as pd
import numpy as np
from scipy import stats
```

2. If you have imported pandas, check that you assign a value to the variable **url**

Q: I can't convert values to int or float Even after running

```
df=df.replace(r'^\s*$', np.nan, regex=True)
```

A: Running the above code only replace whitespace with a nan value

nan stands for Not a Number, and can not be converted to int or float. The reason we convert missing values to nan is that pandas lets us handle those values quite simple. If you want to assign nan's a value, you can use

```
df["Column"] = df["Column"].replace(np.nan, VALUE)
#Column is a placeholder for the column you want to change.
#In this example we have the columns StudentID,Age,email,hrsStudy,FinalGrade
```

After you have replaced the nan values you want, you can drop rows containing nan with:

```
df.dropna(inplace = True)
```

Q: df["FinalGrade"].plot.box() don't work

1. When running this code I get an error
2. The code run, but I can't see a plot

A:

1. If you get an error and your code looks correct, reinstall matplotlib. In your terminal, make sure you are in the **Lab2** folder, then run:

```
uv add --reinstall matplotlib
```

Restart the kernel in your notebook and try again.

2. If your code runs, but only produce

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f7cb044f7d0>
```

add

```
%matplotlib inline
```

in the includes section

Q: How do I remove outliers?

A: Check [Geeksforgeeks](#) for background on the z-score method.

In this lab we use z-scores to find the outliers, then drop those rows:

```
z_scores = stats.zscore(df["FinalGrade"])
abs_z_scores = np.abs(z_scores)
df.drop(df[abs_z_scores > 3].index, inplace=True)
```

This is the approach used in [solution.ipynb](#). **Q: Can you provide some tutorials for jupyter and pandas?**

I need some good tutorials to get me started. Can you recommend any?

A: Yes

If you are new to jupyter notebook and pandas [this youtube video](#) will be useful.

[This site](#) covers many important aspects of pandas, and I use it often as a reference.